Wireless Charging Facility Deployment and Adaptive Battery Configuration under Service-Plan Uncertainty

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Abstract

This abstract will be written after the main body is stable.

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1. Introduction

Electrifying bus fleets requires coordinated planning of charging infrastructure, battery capacity, and the use of limited charging resources under uncertain future service requirements. Wireless charging technologies expand the set of charging opportunities by allowing buses to receive energy at selected stops or road links during regular service. This capability turns infrastructure planning from a stand-alone siting decision into a joint planning problem linking wireless facility deployment, route-level battery configuration, and the use of existing plug-in chargers.

Wireless charging facilities are long-lived assets, whereas transit service plans may change with future demand and procurement planning. A layout that is attractive for one future service plan may perform poorly under another because route frequencies, trip chains, active-fleet requirements, and charging-resource use may change. The infrastructure decision must therefore be made before the future service plan is known, whereas route-level battery configurations and operational decisions can adapt after a service-plan scenario is realized.

The central planning trade-off is shaped by the interaction among wireless infrastructure, battery capacity, and finite plug-in charging capacity. More in-route wireless charging can reduce battery requirements on some routes and lower their residual demand for plug-in charging. At the same time, facility placement changes when and where that residual demand occurs. Because plug-in chargers are limited shared resources, concurrent charging requirements couple route-level decisions that would otherwise be separable.

We capture this timing and coupling structure with a two-stage stochastic mixed-integer planning model. The first stage selects stationary and dynamic wireless charging facilities. The second stage selects route-level battery configurations and, conditional on those selections, determines representative-day charging allocation and energy management for the realized service-plan scenario. The resulting recourse problems contain discrete battery choices, plug-in charger-occupancy decisions, and shared plug-in capacity constraints. Consequently, both the deterministic equivalent and the classical integer L-shaped decomposition become difficult to scale as the network and scenario set grow.

To solve this model, we develop a reformulation-enabled integer L-shaped method. The method differs from a standard integer L-shaped decomposition by changing what recourse information is exposed to the master and by separating inexpensive lower-bound generation from exact recourse verification. This design preserves exactness when the cut loop converges and creates the algorithmic components evaluated in the computational study.

The paper makes three contributions. First, it identifies a layered two-stage stochastic mixed-integer structure in which here-and-now facility decisions interact with scenario-dependent discrete configurations and shared-resource operational recourse. The wireless-charging application provides a concrete instantiation in which conditioning on battery choices reveals route-local operations, limited plug-in capacity couples routes, and wireless facilities and batteries form a joint operational resource order. Second, it develops an exact reformulation-enabled integer L-shaped algorithm that combines scenario-dependent decision variables lifting, a two-phase relaxation-verification scheme, and monotonicity-strengthened cuts for mixed-integer recourse. Third, it provides comprehensive computational evidence through an extensive algorithmic ablation study and systematic analyses of performance with respect to network size, scenario count, and the gap between the Phase 1 lower-bounding relaxation and the exact Phase 2 recourse evaluation.

The experiments use wireless-charging planning instances constructed from Singapore bus-service data for routes operated by Tower Transit. They examine whether the model produces interpretable planning decisions,

how the algorithmic components affect network-scale and scenario-scale performance, and how the complete method responds as shared plug-in capacity makes the Phase 1 relaxation more or less aligned with exact verification.

The remainder of the paper is organized as follows. Section 2 reviews related work on electric-bus charging infrastructure planning and stochastic decomposition methods. Section 3 presents the two-stage formulation and identifies its recourse structure. Section 4 develops the lifted two-phase method and monotonicity-strengthened cuts. Section 5 reports the computational results. Section 6 concludes.

2. Literature Review

This paper intersects two literature streams. The first studies charging infrastructure, battery configuration, and energy-feasible operations for electric buses; the second develops exact decomposition methods for stochastic design problems with mixed-integer recourse.

2.1. Electric-bus charging and wireless infrastructure planning

Electric-bus planning links strategic infrastructure decisions with tactical battery and charging decisions. Charging-site selection and battery capacity determine the resources available to a fleet, while vehicle assignment, charging schedules, and route energy constraints determine whether those resources can support the prescribed service. Integrated optimization models therefore evaluate infrastructure and battery investments together with energy-feasible service operations.

Depot and terminal charging studies focus on the timing of plug-in charging under vehicle schedules, electricity prices, and charger-capacity limits. Opportunity-charging models extend this interaction to charging during service, allowing energy to be replenished between trips or at selected stops. When chargers have finite capacity, station–time occupancy constraints couple buses or routes that would otherwise admit separate energy-feasibility calculations.

Wireless charging introduces a further infrastructure layer. SWC creates charging opportunities at selected stations, whereas DWC supplies energy while a bus traverses an equipped road segment. The associated planning literature examines how the location and extent of these facilities affect battery requirements, route energy feasibility, and investment and charging costs. These models establish the facility–battery trade-off that underlies the present application.

Together, these studies provide the main application components of our model: wireless-facility placement, route-level battery configuration, energy-feasible trip chains, and limited plug-in charging resources. We combine these components under service-plan uncertainty, with wireless facilities selected before the future service plan is known and battery configurations and charging decisions adapted to its realization. This setting links long-term infrastructure planning to shared-resource mixed-integer recourse.

2.2. Decomposition methods for stochastic design problems

Two-stage stochastic models for facility location and network design separate infrastructure choices made before uncertainty from scenario-dependent recourse. Scenario decomposition is particularly attractive when the deterministic equivalent is large, because fixing the first-stage design separates the evaluation of distinct scenarios. The computational challenge increases when the recourse contains discrete configuration or operating decisions rather than a continuous linear program.

Benders decomposition and the L-shaped method use subproblem information to iteratively refine a master representation of future cost and feasibility. For continuous recourse, linear-programming duality supplies this information directly. Mixed-integer recourse instead requires integer L-shaped, logic-based, or related combinatorial mechanisms, and exact scenario evaluation can become the dominant computational burden.

One line of research improves this process by changing the decomposition architecture. Richer master formulations expose selected recourse decisions or partial recourse descriptions, while lower-bounding relaxations and internal subproblem decompositions reduce the cost of scenario evaluation. These choices determine both the strength of the master bound and the computational effort required to generate it.

A complementary line of research strengthens the information returned by the subproblems. Logic-based and combinatorial cuts exploit discrete feasibility structure, while dominance relations and problem-specific valid inequalities can extend a correction beyond the single configuration at which it was generated. Such strengthening depends on a verifiable ordering or implication structure in the underlying recourse model.

These studies show that master representation, subproblem organization, and cut design must be matched to the recourse structure. The setting considered here combines scenario-dependent discrete battery configurations, conditionally route-local operations, and limited shared-resource coupling within each scenario. This layered structure motivates an exact decomposition method designed around their interaction rather than a direct application of scenario decomposition alone.

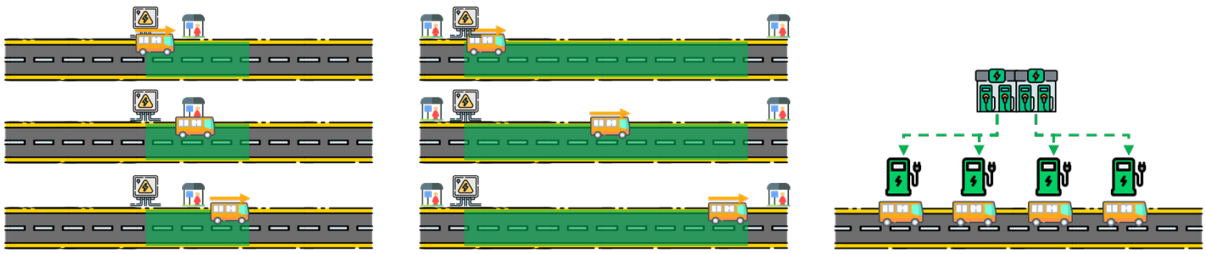
3. Problem Formulation and Recourse Structure

This section formulates the wireless charging infrastructure planning problem as a two-stage stochastic program. The formulation separates long-lived wireless-facility decisions from scenario-adaptive battery and charging decisions, and then identifies the recourse structure used by the algorithm in Section 4.

3.1. Two-stage stochastic formulation

Let Ω be the set of service-plan scenarios, and let p_ω be the probability of scenario $\omega \in \Omega$, where $p_\omega > 0$ and $\sum_{\omega\in\Omega} p_\omega = 1$. Each scenario represents one plausible future service plan and fixes the route frequencies, timetables, bus trip chains, active-fleet requirements, and trip-level energy requirements, while the operated route set $\mathcal{R}$ is common across scenarios. The planner first installs long-lived wireless charging facilities before the future service plan is known. After a scenario is realized, the second-stage recourse selects route-level battery capacities and determines the corresponding representative-day charging allocation and energy management.

Let $\mathcal{S}$ be the station set, $\mathcal{L}$ the directed link set between consecutive stations, and $C \subseteq \mathcal{S}$ the set of existing interchanges or hubs equipped with shared plug-in chargers. Stationary wireless charging (SWC) can be installed at any station in $\mathcal{S} \setminus C$, while dynamic wireless charging (DWC) can be installed on any directed link in $\mathcal{L}$. The first-stage binary variables w_s^{S} and w_l^{D} indicate whether SWC is installed at station $s \in \mathcal{S} \setminus C$ and DWC is installed on link $l \in \mathcal{L}$, respectively. Stations in C have fixed, limited plug-in capacity and are not candidates for SWC deployment. Figure 1 illustrates the three charging resources represented in the formulation.



(a) SWC **(b)** DWC **(c)** Plug-in charging

Figure 1: Charging resources represented in the formulation. (a) SWC is installed near selected stations and supplies energy during low-speed station approaches and departures. (b) DWC is installed on selected directed links and supplies energy during link traversals. (c) Existing plug-in chargers at hubs and interchanges provide limited station–time capacity shared across routes.

The compact two-stage stochastic formulation is

$$\min_{w^{\mathrm{S}},w^{\mathrm{D}}} \quad C^{\mathrm{fac}}(w) + \sum_{\omega\in\Omega} p_\omega \Phi_\omega(w) \tag{1}$$

$$\text{s.t.} \quad w_s^{\mathrm{S}} \in \{0,1\}, \qquad s \in \mathcal{S} \setminus C, \tag{2}$$

$$w_l^{\mathrm{D}} \in \{0,1\}, \qquad l \in \mathcal{L}. \tag{3}$$

where $w = (w^{\mathrm{S}}, w^{\mathrm{D}})$ and

$$C^{\mathrm{fac}}(w) = \sum_{s\in\mathcal{S}\setminus C} c_s^{\mathrm{S}} w_s^{\mathrm{S}} + \sum_{l\in\mathcal{L}} c_l^{\mathrm{D}} w_l^{\mathrm{D}}.$$

Here c_s^{S} and c_l^{D} denote the equivalent daily infrastructure costs, obtained by annualizing the installation costs over the planning horizon and converting them to a representative-day basis. Battery investment costs are converted on the same basis, so that all objective terms are expressed on a common daily scale. The vector w is the nonanticipative here-and-now facility decision, and $\Phi_\omega(w)$ is the optimal second-stage value in scenario ω.

Let z_ω collect the route-level battery selections, x_ω the plug-in charger occupancy decisions, and y_ω the charging amounts and battery energy states. For a fixed facility layout w, the scenario recourse value $\Phi_\omega(w)$ is the minimum scenario cost

$$\Phi_\omega(w) = \min_{z_\omega,x_\omega,y_\omega} \left\{ C_\omega^{\mathrm{bat}}(z_\omega) + C_\omega^{\mathrm{chg}}(y_\omega) : (z_\omega, x_\omega, y_\omega) \in \mathcal{P}_\omega(w) \right\}. \tag{4}$$

Fleet-size adjustment costs and the residual value of the existing batteries are scenario constants and are therefore omitted from $\Phi_\omega(w)$. The next subsection expands $\mathcal{P}_\omega(w)$.

3.2. Scenario recourse formulation

For a fixed scenario ω and route $r \in \mathcal{R}$, let $\mathcal{M}_{\omega,r}$ contain the bus trip chains on route r. Each chain corresponds to one bus duty consisting of an ordered sequence of trips, so $n_{\omega,r} = |\mathcal{M}_{\omega,r}|$ is the route fleet requirement in scenario ω. The sets $\mathcal{V}_{\omega,r}$ and $\mathcal{A}_{\omega,r}$ collect, respectively, all station-visit and directed-traversal event copies generated by these trip chains. Each event copy uniquely identifies its trip chain, trip, and position within the duty, so explicit event-ownership indices are suppressed for notational simplicity. Distinct operational occurrences remain separate even when they correspond to the same physical station or directed link.

Figure 2 illustrates the event-level representation of a representative bus duty and the charging opportunities embedded in its sequence of trips.

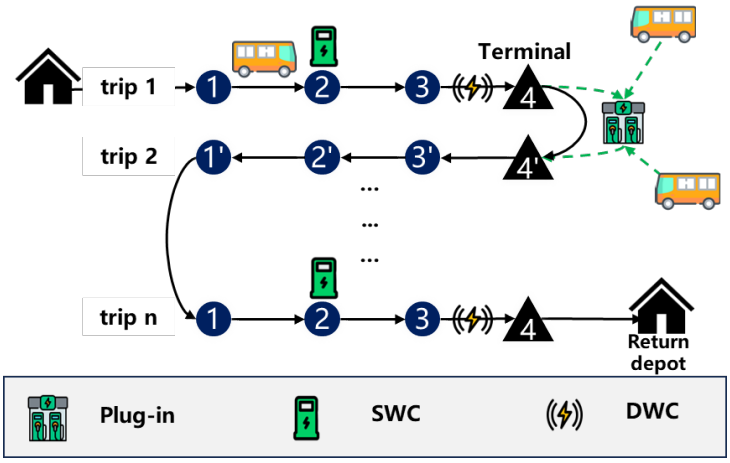


Figure 2: Event-level representation of a representative bus duty and its charging opportunities. A bus sequentially operates multiple trips composed of ordered station visits and directed traversals. SWC opportunities are associated with eligible station visits, DWC opportunities with equipped traversals, and plug-in charging with eligible terminal or hub visits. Limited plug-in chargers may be shared by buses arriving from different trips or routes.

A consolidated notation table is provided in Appendix Appendix A.

Battery configuration and cost.. Battery option $k \in \mathcal{K}$ has capacity β_k, and the scenario-dependent binary variable $z_{\omega,r,k}$ indicates whether it is selected for route r. The route-level battery-choice constraints are

$$\sum_{k\in\mathcal{K}} z_{\omega,r,k} = 1, \qquad r \in \mathcal{R}, \tag{5}$$

$$B_{\omega,r} = \sum_{k\in\mathcal{K}} \beta_k z_{\omega,r,k}, \qquad r \in \mathcal{R}, \tag{6}$$

$$z_{\omega,r,k} \in \{0,1\}, \qquad r \in \mathcal{R},\ k \in \mathcal{K}. \tag{7}$$

Constraints (5)–(7) select one common battery option for all buses assigned to each route and define the induced route battery capacity $B_{\omega,r}$. With representative-day equivalent capacity cost c^{bat}, the scenario battery cost is

$$C^{\text{bat}}_{\omega}(z_\omega) = c^{\text{bat}} \sum_{r\in\mathcal{R}} n_{\omega,r} \sum_{k\in\mathcal{K}} \beta_k z_{\omega,r,k}.$$

Shared plug-in charging.. Plug-in charger use is modeled over scenario time slots $\mathcal{T}_\omega$. For route r, $\mathcal{V}^{\text{P}}_{\omega,r} \subseteq \mathcal{V}_{\omega,r}$ contains the visit event copies eligible for plug-in charging. An eligible visit event v can use plug-in charging only during $\mathcal{T}_v \subseteq \mathcal{T}_\omega$, and $s(v)$ maps it to its physical station. Let q^{P}_v denote the grid-side plug-in energy supplied during visit v. The binary variable $x_{v\tau}$ records whether visit v occupies a plug-in charger in time slot τ; each occupied slot delivers up to κ units of grid-side energy, and station s has capacity U_s. The plug-in charging constraints are

$$0 \le q^{\text{P}}_v \le \kappa \sum_{\tau\in\mathcal{T}_v} x_{v\tau}, \qquad r \in \mathcal{R},\ v \in \mathcal{V}^{\text{P}}_{\omega,r}, \tag{8}$$

$$\sum_{r\in\mathcal{R}} \sum_{\substack{v\in\mathcal{V}^{\text{P}}_{\omega,r}:\\ s(v)=s,\ \tau\in\mathcal{T}_v}} x_{v\tau} \le U_s, \qquad s \in C,\ \tau \in \mathcal{T}_\omega, \tag{9}$$

$$x_{v\tau} \in \{0,1\}, \qquad r \in \mathcal{R},\ v \in \mathcal{V}^{\text{P}}_{\omega,r},\ \tau \in \mathcal{T}_v. \tag{10}$$

Constraints (8)–(10) limit energy delivery during each visit and enforce available plug-in capacity at every station–time pair. Constraint (9) provides the main cross-route coupling within a scenario.

Route-local operations and charging cost.. Let y_ω collect the nonnegative continuous charging and energy-state variables q_v^{P}, q_v^{S}, q_a^{D}, q_m^{N}, e_v^{arr}, and e_v^{dep}. For each traversal event a, $\ell(a)$ denotes its physical directed link, while tail(a) and head(a) denote its departure and arrival visit event copies. At daytime and overnight electricity prices $c^{\mathrm{day}} \geq 0$ and $c^{\mathrm{night}} \geq 0$, the scenario charging cost is

$$C_\omega^{\mathrm{chg}}(y_\omega) = c^{\mathrm{day}} \left(\sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_{\omega,r}^{\mathrm{P}}} q_v^{\mathrm{P}} + \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_{\omega,r}^{\mathrm{S}}} q_v^{\mathrm{S}} + \sum_{r \in \mathcal{R}} \sum_{a \in \mathcal{A}_{\omega,r}} q_a^{\mathrm{D}} \right) + c^{\mathrm{night}} \sum_{r \in \mathcal{R}} \sum_{m \in \mathcal{M}_{\omega,r}} q_m^{\mathrm{N}}.$$

The parameters d_a, d_m^0, and d_m^N represent traversal and depot-boundary energy requirements, while $\underline{\rho}$ and $\overline{\rho}$ are the admissible state-of-charge fractions. Trip-level energy requirements are exogenous and independent of the selected battery capacity; each trip chain starts at the upper state-of-charge limit, and overnight depot charging is sufficient to restore every bus before the next representative day. The route-local operational constraints are

$$0 \leq q_v^{\mathrm{S}} \leq \bar{q}_v^{\mathrm{S}} w_{s(v)}^{\mathrm{S}}, \qquad v \in \mathcal{V}_{\omega,r}^{\mathrm{S}}, \tag{11}$$

$$0 \leq q_a^{\mathrm{D}} \leq \bar{q}_a^{\mathrm{D}} w_{\ell(a)}^{\mathrm{D}}, \qquad a \in \mathcal{A}_{\omega,r}. \tag{12}$$

$$\underline{\rho} B_{\omega,r} \leq e_v^{\mathrm{arr}} \leq \overline{\rho} B_{\omega,r}, \qquad v \in \mathcal{V}_{\omega,r}, \tag{13}$$

$$\underline{\rho} B_{\omega,r} \leq e_v^{\mathrm{dep}} \leq \overline{\rho} B_{\omega,r}, \qquad v \in \mathcal{V}_{\omega,r}, \tag{14}$$

$$e_v^{\mathrm{dep}} = e_v^{\mathrm{arr}} + \eta^{\mathrm{P}} q_v^{\mathrm{P}}, \qquad v \in \mathcal{V}_{\omega,r}^{\mathrm{P}}, \tag{15}$$

$$e_v^{\mathrm{dep}} = e_v^{\mathrm{arr}} + \eta^{\mathrm{S}} q_v^{\mathrm{S}}, \qquad v \in \mathcal{V}_{\omega,r}^{\mathrm{S}}, \tag{16}$$

$$e_{\mathrm{head}(a)}^{\mathrm{arr}} = e_{\mathrm{tail}(a)}^{\mathrm{dep}} - d_a + \eta^{\mathrm{D}} q_a^{\mathrm{D}}, \qquad a \in \mathcal{A}_{\omega,r}, \tag{17}$$

$$e_{v_m^0}^{\mathrm{arr}} = \overline{\rho} B_{\omega,r} - d_m^0, \qquad m \in \mathcal{M}_{\omega,r}, \tag{18}$$

$$e_{v_m^N}^{\mathrm{dep}} - d_m^N \geq \underline{\rho} B_{\omega,r}, \qquad m \in \mathcal{M}_{\omega,r}, \tag{19}$$

$$\eta^{\mathrm{P}} q_m^{\mathrm{N}} = \overline{\rho} B_{\omega,r} - e_{v_m^N}^{\mathrm{dep}} + d_m^N, \qquad m \in \mathcal{M}_{\omega,r}. \tag{20}$$

Constraints (11)–(12) link wireless charging quantities to installed facilities, while constraints (13)–(14) bound the energy states relative to $B_{\omega,r}$. Constraints (15)–(17) propagate energy through charging and traversals, and constraints (18)–(20) close each trip chain.

3.3. Recourse structure

For a fixed facility layout w, the expected recourse term is separable across scenarios. This yields a standard integer L-shaped reference in which each scenario is evaluated as a monolithic mixed-integer recourse problem over $(z_\omega, x_\omega, y_\omega)$; we refer to this direct decomposition as Natural ILS.

Within each scenario, battery and charging costs are additive across routes, and all constraints except constraint (9) involve variables associated with a single route. The latter aggregates plug-in occupancy across routes at each station–time pair. That is, the scenario recourse is conditionally route-separable apart from this localized shared-capacity coupling. The formulation also induces a facility–battery resource order. Because trip-level energy consumption is independent of battery capacity, retaining all installed wireless facilities while adding facilities and/or weakly increasing route battery capacities preserves operational feasibility. Moreover, the minimum charging cost under such a resource-dominating configuration is no greater than that under the original configuration.

The recourse structure therefore combines scenario separability, a route-local block structure with limited shared-capacity coupling, and a facility–battery resource order. Section 4 makes the scenario-dependent battery configuration explicit through an equivalent lifted value-function reformulation. The resulting representation, conditional on (w, z_ω), supports both route-level lower bounding and resource-monotonicity strengthening.

4. Reformulation-Enabled Integer L-Shaped Method

This section develops an exact integer L-shaped method for the two-stage formulation. The method has three main components: a lifted value-function reformulation, a two-phase separation scheme, and resource-monotonicity-strengthened cuts.

4.1. Lifted value-function reformulation

In a natural integer L-shaped decomposition of (1), only the facility variables w are explicit among the discrete configuration decisions, while each scenario recourse evaluation jointly optimizes the battery configuration z_ω and the operational decisions (x_ω, y_ω). This aggregation keeps the battery choice embedded in the mixed-integer recourse problem. Consequently, the master-side support of the integer L-shaped cuts is restricted to the facility variables.

We therefore make z_ω explicit in the lifted decomposition. The facility layout w remains a here-and-now decision, whereas z_ω and (x_ω, y_ω) remain scenario-adaptive decisions made after scenario ω is realized. Thus, lifting changes which variables are visible to the decomposition master, but it does not change the information structure.

To make this separation explicit, we first condition on a candidate facility–battery configuration. For a fixed (w, z_ω), the route battery capacities are fixed through (6), and the remaining scenario evaluation is the exact operational MIP

$$\begin{aligned} Q_\omega(w, z_\omega) = \min_{x_\omega, y_\omega} \quad & C^{\mathrm{chg}}_\omega(y_\omega) \\ \text{s.t.} \quad & (8)\text{–}(10), \\ & (11)\text{–}(20), \quad r \in \mathcal{R}. \end{aligned} \tag{21}$$

The first constraint group retains scenario-level plug-in occupancy and shared-capacity constraints, while the second contains all route-indexed operational constraints. If this MIP is infeasible, we set $Q_\omega(w, z_\omega) = +\infty$.

With Q_ω defined in this way, the scenario value in (4) satisfies

$$\begin{aligned} \Phi_\omega(w) = \min_{z_\omega} \quad & C^{\mathrm{bat}}_\omega(z_\omega) + Q_\omega(w, z_\omega) \\ \text{s.t.} \quad & (5)\text{–}(7). \end{aligned} \tag{22}$$

Let $z = (z_\omega)_{\omega\in\Omega}$ collect the scenario-dependent battery-selection vectors. Substituting (22) into the original two-stage formulation (1) yields the following exact lifted value-function formulation:

$$\begin{aligned} V^L = \min_{w,z} \quad & \left[C^{\mathrm{fac}}(w) + \sum_{\omega\in\Omega} p_\omega C^{\mathrm{bat}}_\omega(z_\omega) \right] + \sum_{\omega\in\Omega} p_\omega Q_\omega(w, z_\omega) \\ \text{s.t.} \quad & (2)\text{–}(3), \\ & (5)\text{–}(7), \qquad\qquad \omega \in \Omega. \end{aligned} \tag{23}$$

In this formulation, both the facility variables w and the scenario-dependent battery variables z_ω are explicit in the decomposition, whereas the remaining operational decisions are represented through the conditional value function $Q_\omega(w, z_\omega)$. The lifted formulation is exact because (22) simply rewrites each scenario recourse value as the sum of battery cost and the exact conditional operational value. It changes the decomposition structure but not the decision timing, information structure, feasible first-stage decisions, or optimal value of (1).

4.2. Route-level lower-bounding relaxation and separation

The lifted reformulation makes the scenario battery variables explicit, but exact lifted integer L-shaped separation remains computationally demanding. For each candidate (w, z_ω), exact scenario evaluation must solve an operational MIP with route coupling from shared plug-in capacity and binary plug-in occupancy. The route-level lower-bounding procedure developed in this section constitutes Phase 1 of the proposed method.

Partition the operating vector by route as $y_\omega = (y_{\omega,r})_{r\in\mathcal{R}}$, with $C^{\mathrm{chg}}_\omega(y_\omega) = \sum_{r\in\mathcal{R}} C^{\mathrm{chg}}_{\omega,r}(y_{\omega,r})$. The plug-in charging block (8)–(10) comprises charging-limit, shared-capacity, and occupancy-domain constraint families. Phase 1 removes the shared station–time capacity family (9) and equivalently replaces the other two families with

$$0 \le q^{\mathrm{P}}_v \le \kappa |\mathcal{T}_v|, \qquad v \in \mathcal{V}^{\mathrm{P}}_{\omega,r}. \tag{24}$$

The resulting relaxation separates by route. For a fixed route configuration $(w, z_{\omega,r})$, all constraints in the following LP are imposed for route r. The route-level LP value is

$$\begin{aligned} Q^R_{\omega,r}(w, z_{\omega,r}) = \min_{y_{\omega,r}} \quad & C^{\mathrm{chg}}_{\omega,r}(y_{\omega,r}) \\ \text{s.t.} \quad & (11)\text{–}(20), \\ & (24). \end{aligned} \tag{25}$$

We set $Q^R_{\omega,r}(w, z_{\omega,r}) = +\infty$ if this route LP is infeasible. Let $Q^R_\omega(w, z_\omega) = \sum_{r\in\mathcal{R}} Q^R_{\omega,r}(w, z_{\omega,r})$ denote the aggregate Phase 1 value for scenario ω. Substituting Q^R_ω for Q_ω in (23) gives the Phase 1 lower-bounding formulation:

$$\begin{aligned} V^R = \min_{w,z} \quad & \left[C^{\text{fac}}(w) + \sum_{\omega\in\Omega} p_\omega C^{\text{bat}}_\omega(z_\omega)\right] + \sum_{\omega\in\Omega} p_\omega Q^R_\omega(w, z_\omega) \\ \text{s.t.} \quad & (2)\text{–}(3), \\ & (5)\text{–}(7), \qquad\qquad \omega \in \Omega. \end{aligned} \tag{26}$$

Proposition 1 (Validity of the Phase 1 relaxation)**.** *For any lifted design configuration (w, z_ω) satisfying the design and battery-selection constraints,*

$$Q^R_\omega(w, z_\omega) \le Q_\omega(w, z_\omega). \tag{27}$$

If any route-level Phase 1 LP is infeasible, the exact operational subproblem is infeasible.

Proof. If the exact operational problem is infeasible, then $Q_\omega(w, z_\omega) = +\infty$ and (27) holds by convention. Otherwise, consider any feasible exact operational solution (x_ω, y_ω). Constraints (8) and (10) imply that its plug-in charging amounts satisfy (24). All remaining constraints are unchanged and route-local; hence, the route-specific components $y_{\omega,r}$ form a feasible solution to (25) for every route r. Because the charging cost is route additive, the aggregate cost of these route solutions equals $C^{\text{chg}}_\omega(y_\omega)$. Minimizing over the route LPs and then over the exact operational solutions gives $Q^R_\omega(w, z_\omega) \le Q_\omega(w, z_\omega)$. The same construction shows that exact feasibility requires every route LP to be feasible, proving the final statement. □

The converse does not hold in general: feasibility of all route LPs does not guarantee exact scenario feasibility because Phase 1 omits the shared plug-in capacity constraints. Because (26) and (23) have the same lifted design domain, Proposition 1 implies $V^R \le V^L$.

To solve the Phase 1 formulation by integer L-shaped separation, introduce an epigraph proxy $\theta^R_{\omega,r}$ for each route value $Q^R_{\omega,r}(w, z_{\omega,r})$. Let $\mathcal{F}^R_{\omega,r}$ and $\mathcal{O}^R_{\omega,r}$ denote the current route-level feasibility and optimality cut pools, and write $\theta^R = \{\theta^R_{\omega,r}\}_{\omega\in\Omega, r\in\mathcal{R}}$. The Phase 1 restricted master is

$$\begin{aligned} \min_{w,z,\theta^R} \quad & \left[C^{\text{fac}}(w) + \sum_{\omega\in\Omega} p_\omega C^{\text{bat}}_\omega(z_\omega)\right] + \sum_{\omega\in\Omega} p_\omega \sum_{r\in\mathcal{R}} \theta^R_{\omega,r} \\ \text{s.t.} \quad & (2)\text{–}(3), \\ & (5)\text{–}(7), && \omega \in \Omega, \\ & \theta^R_{\omega,r} \ge 0, && \omega \in \Omega,\ r \in \mathcal{R}, \\ & \psi^R_{\omega,r}(w, z_{\omega,r}) \ge 0, && \omega \in \Omega,\ r \in \mathcal{R},\ \psi^R_{\omega,r} \in \mathcal{F}^R_{\omega,r}, \\ & \theta^R_{\omega,r} \ge \varphi^R_{\omega,r}(w, z_{\omega,r}), && \omega \in \Omega,\ r \in \mathcal{R},\ \varphi^R_{\omega,r} \in \mathcal{O}^R_{\omega,r}. \end{aligned} \tag{28}$$

Each feasible route LP yields a dual optimality cut in $(w, z_{\omega,r}, \theta^R_{\omega,r})$, whereas an infeasible route LP yields a Farkas feasibility cut. Appendix Appendix B.1 gives their explicit forms.

By Proposition 1, Phase 2 can retain the Phase 1 lower bound, Phase 1 route proxies and accumulated Phase 1 cut pools.

4.3. Resource monotonicity and strengthened scenario cuts

We first establish a scenario-level resource order and then use it to strengthen the pointwise integer L-shaped cuts employed in exact scenario separation. As in (21), set $Q_\omega(w, z_\omega) = +\infty$ when the exact operational MIP is infeasible. The argument uses the model specification that trip-level energy requirements do not vary with the selected battery capacity.

Resource order and monotonicity. Let $\mathcal{W}$ index all candidate SWC stations and DWC links, and let w_j denote the corresponding binary facility variable. Fixed plug-in charger capacities are not included in this resource order. For route r, write $B_{\omega,r}(z_{\omega,r}) = \sum_{k\in\mathcal{K}} \beta_k z_{\omega,r,k}$. For two scenario configurations satisfying the design constraints (2)–(3) and battery-selection constraints (5)–(7), write $(w, z_\omega) \preceq_\omega (\bar{w}, \bar{z}_\omega)$ if

$$w_j \le \bar{w}_j, \quad j \in \mathcal{W}, \qquad B_{\omega,r}(z_{\omega,r}) \le B_{\omega,r}(\bar{z}_{\omega,r}), \quad r \in \mathcal{R}.$$

Proposition 2 (Scenario-level resource monotonicity)**.** *For a fixed scenario $\omega \in \Omega$, if $(w, z_\omega) \preceq_\omega (\bar{w}, \bar{z}_\omega)$, then*

$$Q_\omega(\bar{w}, \bar{z}_\omega) \le Q_\omega(w, z_\omega).$$

Proof. If $Q_\omega(w, z_\omega) = +\infty$, the inequality is immediate. Otherwise, take any feasible exact operating solution (x_ω, y_ω) under (w, z_ω) and keep all occupancy and charging decisions unchanged. The plug-in constraints remain feasible, and the SWC and DWC charging bounds remain valid because the corresponding facility variables weakly increase. For each route r, write $B_{\omega,r} = B_{\omega,r}(z_{\omega,r})$, $\bar{B}_{\omega,r} = B_{\omega,r}(\bar{z}_{\omega,r})$, and $\Delta_r = \bar{B}_{\omega,r} - B_{\omega,r} \geq 0$. Increase every arrival and departure energy state on that route by $\overline{\rho}\Delta_r$. This common shift preserves all station and link energy-balance equations. Moreover, if $\underline{\rho} B_{\omega,r} \leq e \leq \overline{\rho} B_{\omega,r}$, then, since $\overline{\rho} \geq \underline{\rho}$,

$$\underline{\rho}\bar{B}_{\omega,r} \leq e + \overline{\rho}\Delta_r \leq \overline{\rho}\bar{B}_{\omega,r}.$$

The initial boundary equality is preserved exactly, while the return lower bound remains valid because its left-hand side increases by $\overline{\rho}\Delta_r$ and its right-hand side by $\underline{\rho}\Delta_r$. It also preserves the overnight balance because both $\overline{\rho}B_{\omega,r}$ and the terminal departure energy increase by $\overline{\rho}\Delta_r$.

Hence the transformed solution is feasible under $(\bar{w}, \bar{z}_\omega)$ and has the same operational objective value. Minimization therefore gives $Q_\omega(\bar{w}, \bar{z}_\omega) \leq Q_\omega(w, z_\omega)$. □

This monotonicity concerns Q_ω only and does not imply monotonicity of $C^{\text{bat}}_\omega(z_\omega) + Q_\omega(w, z_\omega)$, because larger batteries may have higher battery cost.

Corollary 1 (Scenario-level dominance implications)**.** *For any two scenario configurations* (w, z_ω) *and* (w^*, z^*_ω) *satisfying the design and battery-selection constraints, Proposition 2 implies:*

$$Q_\omega(w^*, z^*_\omega) = +\infty,\ (w, z_\omega) \preceq_\omega (w^*, z^*_\omega) \Rightarrow Q_\omega(w, z_\omega) = +\infty, \tag{29}$$

$$Q_\omega(w^*, z^*_\omega) < +\infty,\ (w, z_\omega) \preceq_\omega (w^*, z^*_\omega) \Rightarrow Q_\omega(w, z_\omega) \geq Q_\omega(w^*, z^*_\omega). \tag{30}$$

Pointwise and strengthened cuts. Consider an evaluated scenario configuration (w^*, z^*_ω). Let $Q^*_\omega = Q_\omega(w^*, z^*_\omega)$ denote its exact operational value, and let $k^*_{\omega,r}$ be the selected battery option on route r. Because charging costs are nonnegative, we use $L_\omega = 0$ as a global lower bound and impose $\theta_\omega \geq L_\omega$. The pointwise mismatch expression is

$$D_\omega(w, z_\omega; w^*, z^*_\omega) = \sum_{j \in \mathcal{W}: w^*_j = 0} w_j + \sum_{j \in \mathcal{W}: w^*_j = 1} (1 - w_j) + \sum_{r \in \mathcal{R}} \left(1 - z_{\omega,r,k^*_{\omega,r}}\right).$$

The pointwise scenario cuts are

$$\begin{aligned} D_\omega(w, z_\omega; w^*, z^*_\omega) &\geq 1, && \text{if } Q^*_\omega = +\infty, \\ \theta_\omega &\geq Q^*_\omega - (Q^*_\omega - L_\omega) D_\omega(w, z_\omega; w^*, z^*_\omega), && \text{if } Q^*_\omega < +\infty. \end{aligned}$$

Under the binary facility domains and the route-level battery-choice constraints, $D_\omega = 0$ if and only if $(w, z_\omega) = (w^*, z^*_\omega)$; otherwise $D_\omega \geq 1$, yielding the standard pointwise feasibility and optimality corrections.

Corollary 1 extends these pointwise corrections to dominated resource configurations. Define the scenario-level monotonicity mismatch expression

$$I_\omega(w, z_\omega; w^*, z^*_\omega) = \sum_{j \in \mathcal{W}: w^*_j = 0} w_j + \sum_{r \in \mathcal{R}} \sum_{k \in \mathcal{K}: \beta_k > \beta_{k^*_{\omega,r}}} z_{\omega,r,k}.$$

The monotonicity-strengthened scenario cuts are

$$\begin{aligned} I_\omega(w, z_\omega; w^*, z^*_\omega) &\geq 1, && \text{if } Q^*_\omega = +\infty, \\ \theta_\omega &\geq Q^*_\omega - (Q^*_\omega - L_\omega) I_\omega(w, z_\omega; w^*, z^*_\omega), && \text{if } Q^*_\omega < +\infty. \end{aligned}$$

Under the binary facility domains and the route-level battery-choice constraints, I_ω is a nonnegative integer. Moreover, $I_\omega = 0$ implies $(w, z_\omega) \preceq_\omega (w^*, z^*_\omega)$. If $Q^*_\omega = +\infty$, Corollary 1 therefore implies that every configuration with $I_\omega = 0$ is infeasible, proving the strengthened feasibility cut. If $Q^*_\omega < +\infty$ and $I_\omega = 0$, the same corollary gives $Q_\omega(w, z_\omega) \geq Q^*_\omega$. For $I_\omega \geq 1$, the feasibility cut is satisfied and

$$Q^*_\omega - (Q^*_\omega - L_\omega) I_\omega \leq L_\omega,$$

so the optimality cut is valid under $\theta_\omega \geq L_\omega$. Since $I_\omega \leq D_\omega$ and $Q^*_\omega - L_\omega \geq 0$, the strengthened feasibility and optimality cuts weakly dominate their pointwise counterparts.

4.4. Complete procedure and convergence

Let LB^R and UB^R denote the current lower and upper bounds on the Phase 1 formulation (26). For a prescribed phase-switch tolerance $\varepsilon_1 \geq 0$, Phase 1 switches when

$$g^R = \frac{UB^R - LB^R}{\max\{1, |UB^R|\}} \leq \varepsilon_1.$$

Set $g^R = +\infty$ until a finite UB^R is available. By Proposition 1, LB^R is a valid lower bound on both the Phase 1 formulation and the exact lifted formulation, whereas UB^R applies only to the Phase 1 relaxation and is not an exact incumbent bound. The tolerance ε_1 controls the computational effort devoted to Phase 1 without affecting the validity of Phase 2.

At the phase transition, Phase 2 retains the Phase 1 route proxies and route-level cut pools and introduces a scenario-level proxy θ_ω for the exact operational value $Q_\omega(w, z_\omega)$. The linking constraint $\theta_\omega \geq \sum_{r\in\mathcal{R}} \theta^R_{\omega,r}$ transfers the retained route-level lower bounds to the scenario level. Let $\mathcal{F}_\omega$ denote the Phase 2 scenario-level feasibility cut pool, including RSP and exact scenario feasibility cuts, and let O_ω denote the corresponding optimality cut pool. The Phase 2 restricted master is

$$\begin{aligned}
\min_{w,z,\theta^R,\theta} \quad & \left[C^{\text{fac}}(w) + \sum_{\omega\in\Omega} p_\omega C^{\text{bat}}_\omega(z_\omega)\right] + \sum_{\omega\in\Omega} p_\omega \theta_\omega \\
\text{s.t.} \quad & (2)\text{–}(3), \\
& (5)\text{–}(7), && \omega \in \Omega, \\
& \theta^R_{\omega,r} \geq 0, && \omega \in \Omega,\ r \in \mathcal{R}, \\
& \psi^R_{\omega,r}(w, z_{\omega,r}) \geq 0, && \omega \in \Omega,\ r \in \mathcal{R},\ \psi^R_{\omega,r} \in \mathcal{F}^R_{\omega,r}, \\
& \theta^R_{\omega,r} \geq \varphi^R_{\omega,r}(w, z_{\omega,r}), && \omega \in \Omega,\ r \in \mathcal{R},\ \varphi^R_{\omega,r} \in O^R_{\omega,r}, \\
& \theta_\omega \geq \sum_{r\in\mathcal{R}} \theta^R_{\omega,r}, && \omega \in \Omega, \\
& \theta_\omega \geq L_\omega, && \omega \in \Omega, \\
& \psi_\omega(w, z_\omega) \geq 0, && \omega \in \Omega,\ \psi_\omega \in \mathcal{F}_\omega, \\
& \theta_\omega \geq \varphi_\omega(w, z_\omega), && \omega \in \Omega,\ \varphi_\omega \in O_\omega.
\end{aligned} \tag{31}$$

Before invoking exact MIP separation, each Phase 2 master candidate (w, z, θ^R, θ) is screened scenario by scenario using a scenario-level relaxed subproblem (RSP). For each feasible RSP, violated RSP optimality cuts are added, whereas an infeasible RSP certifies infeasibility of the corresponding exact operational MIP and yields a valid scenario-level feasibility cut before exact separation. When the RSPs are feasible, the algorithm proceeds to exact scenario separation.

If the exact operational MIP is feasible and solved to optimality for every scenario at a restricted-master candidate (w, z), the corresponding feasible objective value is

$$UB^{\text{cand}}(w, z) = C^{\text{fac}}(w) + \sum_{\omega\in\Omega} p_\omega C^{\text{bat}}_\omega(z_\omega) + \sum_{\omega\in\Omega} p_\omega Q_\omega(w, z_\omega). \tag{32}$$

The incumbent is updated as $UB \leftarrow \min\{UB, UB^{\text{cand}}(w, z)\}$, and the optimal value of the current Phase 2 restricted master is a valid lower bound LB. For a finite incumbent UB, define the gap as $g = (UB - LB)/\max\{1, |UB|\}$. Set $g = +\infty$ until a finite incumbent is available. Algorithm 1 summarizes the complete procedure for a final tolerance $\varepsilon \geq 0$.

Theorem 1 (Finite convergence under exact separation)**.** *Assume that $p_\omega > 0$ for all $\omega \in \Omega$, the lifted formulation has a finite optimum, and the binary facility–battery configuration space is finite. Suppose Phase 1 switches after finitely many iterations, all generated cuts are valid and retained, auxiliary RSP separation terminates finitely at each master candidate, and every restricted master and exact Phase 2 scenario MIP is solved to proven infeasibility or optimality. With $\varepsilon = 0$, Algorithm 1 terminates finitely with an optimal solution to the lifted formulation and hence to the original two-stage formulation.*

The proof is provided in Appendix Appendix B.

For $\varepsilon > 0$, the procedure returns the incumbent and valid bounds once $g \leq \varepsilon$; if no finite incumbent has been obtained, $g = +\infty$ and gap-based termination is not declared.

Input : Scenario data, candidate facilities and batteries, phase-switch tolerance ε_1, final tolerance ε
Output: Facility layout w, scenario battery configurations z, lower bound LB, upper bound UB

Initialize cut pools; set $LB^R, LB \leftarrow -\infty$ and $UB^R, UB \leftarrow +\infty$;

`// Phase 1: route-level lower-bounding separation`
repeat
  Solve (28) to optimality, set LB^R to its optimal value, and update $LB \leftarrow \max\{LB, LB^R\}$;
  Solve route-level LP relaxations for each scenario at the current (w, z_ω);
  Add violated route-level dual optimality cuts for feasible route LPs;
  if *one or more route LPs are infeasible* **then**
    Add violated route-level Farkas cuts and continue;
  end
  Set $UB^R \leftarrow \min\{UB^R,$ evaluated Phase 1 candidate value$\}$;
until $g^R \le \varepsilon_1$;
Transfer LB, route proxies, and route cut pools; initialize (31);

`// Phase 2: exact mixed-integer verification`
repeat
  Solve the restricted master (31) to optimality and set LB to its optimal value;
  Solve the relaxed subproblems (RSPs) for all scenarios;
  Add violated RSP optimality cuts for feasible RSPs;
  if *one or more RSPs are infeasible* **then**
    Add the corresponding RSP feasibility cuts;
    continue to the next master solve;
  end
  Solve the exact scenario operational MIPs to proven infeasibility or optimality;
  if *one or more scenarios are proven infeasible* **then**
    Add the pointwise or strengthened scenario feasibility cuts defined above;
    continue to the next master solve;
  end
  if *all scenario MIPs are proven optimal* **then**
    Compute UB^{cand} using (32);
    Set $UB \leftarrow \min\{UB, UB^{\text{cand}}\}$ and update the incumbent if improved;
    foreach *scenario whose exact recourse is underestimated by* θ_ω **do**
      Add the pointwise or strengthened scenario optimality cuts defined above;
    end
  end
until $g \le \varepsilon$;
return *Best incumbent, LB, and UB*;

Algorithm 1: Two-phase integer L-shaped method

5. Computational Results

This section evaluates the proposed reformulation and solution method on Tower-derived wireless charging planning instances. The experiments use the bus network as a realistic source of shared stops, shared links, and route-level service plans, while the main purpose of the section is to validate algorithmic performance.

5.1. Experimental design and instance families

The test instances are derived from the Tower Transit bus network in Singapore's Jurong area. The resulting data set provides a realistic spatial structure with shared stops, shared road links, and multiple routes that may use the same wireless charging infrastructure. We use these instances as a computational test bed rather than as a public benchmark.

Each scenario represents a plausible future service plan for the same physical network, not random same-day scheduling noise. Scenario-dependent data include route frequencies, timetables, trip chains, active-fleet requirements, trip-level energy requirements, and charging opportunities induced by the corresponding service plan. The first-stage wireless charging facilities are shared across all scenarios, whereas route-level battery configurations, charging allocation, and energy management adapt after the service plan is realized.

We use a compact instance code of the form `N-k`, where `N` indicates the network scale and k indicates the number of service-plan scenarios. Specifically, `S`, `M`, `B`, and `L` denote the small, medium, base, and large Tower-derived networks, respectively. The capacity-coupling experiment appends `Cq` to the base 10-scenario instance, where q denotes the number of existing plug-in chargers at each plug-in location. The network-scale family fixes the scenario count at three and varies the physical network scale. The scenario-count family fixes the base network and varies the number of service-plan scenarios. Motivated by the relaxation-verification structure in Section 4.2, the capacity-coupling family varies plug-in charger capacity on `B-10` to change how strongly residual plug-in charging demand couples route-level recourse decisions. Table 1 summarizes the resulting instance families and their experimental roles.

Travel times are computed from link distances using a nominal operating speed of 30 km/h and then rounded to the one-minute scheduling grid.[1] The remaining parameters specify battery costs, charging rates, charging efficiencies, electricity prices, facility costs, and state-of-charge limits. Table 2 reports the main parameter values used in the computational experiments.

The benchmark ladder consists of five methods: the direct extensive-form MILP, baseline integer L-shaped decomposition, lifted integer L-shaped decomposition, lifted two-phase method, and the full method. The direct extensive-form MILP optimizes the deterministic equivalent with a commercial solver. The baseline integer L-shaped decomposition applies the classical scenario decomposition induced by the original two-stage model. The lifted integer L-shaped decomposition uses the lifted master representation without the two-phase relaxation. Lifted two-phase adds the Phase 1 lower-bounding and Phase 2 verification structure. The full implementation further adds tailored Phase 1 cuts and scenario-level monotonicity-strengthened cuts. Two targeted variants support cut ablations: lifted monotone adds the monotonicity cuts to the lifted single-phase method, while full without Phase 1 custom cuts isolates the effect of monotonicity strengthening under the two-phase framework.

For the decomposition methods, Phase 1 switches at $\varepsilon_1 = 10^{-4}$ (0.01%). Restricted masters and exact scenario MIPs are solved with zero solver gap, and route-level Phase 1 LPs are solved to optimality. The main performance runs use a 7200-second time limit.

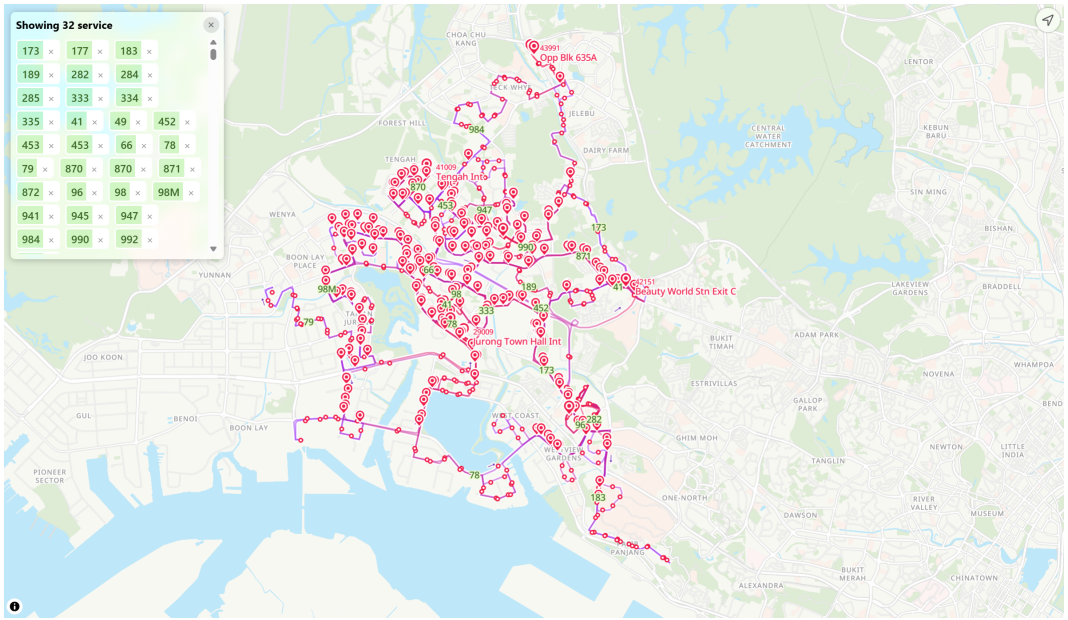


Figure 3: Spatial layout of the base Tower-derived network in the Jurong area.

Table 1: Main instance families and experimental use

Instance family	Data source or variant	Varied dimension	Primary use	Compared methods
`S-3`, `M-3`, `B-3`, `L-3`	`tower_small`, `tower_medium`, `tower`, `tower_large`	Network scale, with three scenarios fixed	Network-scale ablation	Direct MILP, baseline integer L-shaped, lifted integer L-shaped, lifted two-phase, full method
`B-`k, $k \in \{3, 5, 10, 20, 40\}$	`tower`, `tower_5`, `tower_10`, `tower_20`, `tower_40`	Scenario count, with base network fixed	Scenario-count scalability	Full method
`B-10-Cq`, $q \in \{2, 3, 4, 5, 6\}$	Capacity variants of `tower_10`	Existing plug-in charger units, with base network and 10 scenarios fixed	Relaxation-gap stress test	Full method

`B-3` is the representative base instance used for solution interpretation; larger q indicates more existing plug-in charger units.

5.2. Solution characteristics on the base Tower instance

This subsection will use the full-method solution for `B-3` to examine whether the optimized plan is interpretable at the facility, battery, and charging levels. Table 3 lists the outputs to be reported once the final decision data are extracted.

A second compact table will list representative selected stations or links, their facility type, and the routes and scenarios that use them. The accompanying analysis will distinguish shared network-level deployments from route-specific charging opportunities and will relate the selected battery capacities to residual plug-in charging demand.

[1] The one-minute lower bound avoids zero-duration movements on short links and absorbs acceleration, deceleration, signal delay, and station-approach time.

Table 2: Parameter settings for the computational experiments

Parameter	Description	Value	Source
Battery and state of charge			
c^{bat}	Daily-equivalent battery capacity cost	0.148 USD/kWh/day	BloombergNEF price amortized over Wang et al. battery life (BloombergNEF, 2025; Wang et al., 2024)
$\mathcal{K}$	Candidate route-level battery options	10-kWh increments within instance bounds	Case-data preprocessing
$\underline{\rho}, \overline{\rho}$	Minimum and maximum state-of-charge fractions	0.2, 0.8	Wireless bus planning literature (Li et al., 2024)
Charging operation			
κ	Maximum plug-in energy per occupied one-minute slot	2.5 kWh/slot	Najafi et al. (2025)
$\bar{q}_v^{\mathrm{S}}$	Visit-specific SWC opportunity limit	1.0 kWh/min times eligible dwell time	Wang et al. (2024)
$\bar{q}_a^{\mathrm{D}}$	Traversal-specific DWC opportunity limit	1.0 kWh/min times traversal time	Wang et al. (2024)
η^{P}	Plug-in charging efficiency	0.90	Li et al. (2024)
η^{S}	Stationary wireless charging efficiency	0.85	Sun et al. (2024)
η^{D}	Dynamic wireless charging efficiency	0.80	Sun et al. (2024)
U_s	Existing plug-in charger units at plug-in location s	4 chargers in the base setting	Case-study assumption
Infrastructure and electricity cost			
T^{fac}	Facility amortization horizon	30 years	Wang et al. (2024)
c_s^{S}	Daily-equivalent cost of an SWC facility	2.16 USD/day	Wang et al. (2024)
c_l^{D}	Daily-equivalent cost of a DWC facility on link l	9 USD/km/day times link length, plus 0.90 USD/day	Wang et al. (2024)
c^{day}	Daytime electricity price	0.1922 USD/kWh	Singapore tariff data (Singapore Department of Statistics, 2026)
c^{night}	Nighttime electricity price	0.1233 USD/kWh	Singapore tariff data (Singapore Department of Statistics, 2026)

Table 3: Planned summary of the representative `B-3` solution

Result block	Reported quantities	Purpose
Cost composition	Facility, expected battery, daytime charging, and overnight charging costs	Verify how the objective is distributed across planning and recourse decisions
Facility deployment	Numbers of selected SWC stations and DWC links, together with representative selected locations	Identify where the optimized layout creates charging opportunities
Battery adaptation	Route-level battery capacities by scenario and their minimum, median, and maximum values	Show how battery configurations adapt across service plans
Charging allocation	Plug-in, SWC, DWC, and overnight charging energy, plus peak plug-in utilization	Describe how wireless deployment and shared plug-in capacity interact

5.3. Network-scale algorithmic ablation

The network-scale ablation fixes the number of service-plan scenarios at three and increases the physical network size from S-3 to L-3. Table 4 reports the available runs across the core benchmark ladder and tracks how the solvable boundary changes as the lifted and two-phase components are introduced.

Table 4: Network-scale ablation under three service-plan scenarios

Instance	Direct MILP	Baseline integer L-shaped	Lifted integer L-shaped	Lifted two-phase	Full method
S-3	Opt., 4.2s	TL, 98.37%	TL, 77.84%	Opt., 0.4s	Opt., 0.4s
M-3	TL, 0.27%	n.r.	n.r.	Opt., 25.8s	Opt., 25.7s
B-3	n.r.	n.r.	n.r.	Opt., 29.4s	Opt., 32.2s
L-3	n.r.	n.r.	n.r.	TL, 0.092%	TL, 0.092%

Opt. = reached the prescribed optimality tolerance; TL = 7200s time limit; n.r. = not reported in the current run set.
Percentages are final relative gaps at termination.

The results identify the two-phase structure as the main computational transition in this run set. On S-3, the direct MILP is still easy, but the standard integer L-shaped variants make little progress within the time limit because their master lower bounds remain weak. Once route-level lower bounding and exact Phase 2 verification are introduced, both two-phase variants solve S-3, M-3, and B-3. Both two-phase variants reach the time limit on L-3; Table 5 reports their terminal bounds.

Table 5: Two-phase boundary diagnostics on the largest three-scenario instance

Method	Status	UB	LB	Gap (%)	Time (s)
Lifted two-phase	TL	27251.656	27226.537	0.092	7217.5
Full method	TL	27251.656	27226.537	0.092	7231.9

Both runs stop during Phase 2.

The identical terminal bounds place the current network-scale boundary between B-3 and L-3 and show no additional benefit from the full-method enhancements on this instance. Exact Phase 2 verification remains the limiting operation at this scale.

5.4. Scenario-count scalability

This subsection uses the B-k family, with $k \in \{3, 5, 10, 20, 40\}$, to test scenario-count scalability on the base Tower-derived network. To isolate growth in the scenario dimension while holding the network and algorithm fixed, Table 6 reports the full method and separates Phase 1 and Phase 2 time.

Table 6: Scenario-count scalability on the base Tower-derived network

Instance	UB	LB	Gap (%)	P1 time (s)	P2 time (s)	Total (s)
B-3	7869.832	7869.814	0.000	19.2	13.1	32.2
B-5	7852.604	7852.340	0.003	27.4	30.3	57.7
B-10	7862.021	7861.809	0.003	59.0	60.6	119.7
B-20	7850.547	7850.375	0.002	126.6	120.8	247.4
B-40	7849.108	7848.937	0.002	345.5	580.9	926.4

All five runs converge within the prescribed tolerance; reported times are wall-clock seconds.

The full method solves all tested scenario counts within the prescribed tolerance. Total solution time increases from 32.2 seconds for B-3 to 926.4 seconds for B-40, while all final gaps remain below 0.004%. Both phases grow with the scenario count, and Phase 2 becomes the larger time component at 40 scenarios because each master candidate requires more exact scenario evaluations.

5.5. Capacity-coupling and relaxation-gap stress test

This subsection uses B-10-C2 through B-10-C6 to test the effect of shared plug-in charger capacity on the relaxation-verification structure. All five variants are built from B-10; only the number of existing plug-in chargers

is varied. Tighter capacity increases cross-route competition that is absent from the Phase 1 relaxation in Proposition 1, whereas looser capacity should bring the relaxed and exact recourse problems closer. Table 7 reports the full method only and uses the initial Phase 2 gap as a transition diagnostic for the complete algorithm.

Table 7: Capacity-coupling and relaxation-gap stress test on B-10

Instance	Status	UB	LB	Final gap (%)	Initial P2 gap (%)	Total (s)
C2	TL	–	–	–	–	7243.1
C3	Opt.	7871.295	7870.696	0.008	0.676	280.5
C4	Opt.	7862.021	7861.809	0.003	0.562	148.2
C5	Opt.	7859.664	7859.497	0.002	0.532	95.9
C6	Opt.	7859.271	7859.164	0.001	0.527	114.0

Cq denotes q existing plug-in chargers at each plug-in charging location. C2 reaches the Phase 2 time limit before a verified objective pair is recorded.

As plug-in capacity increases from C3 to C6, the initial Phase 2 gap falls from 0.676% to 0.527%, and the final optimality gaps after convergence remain below 0.008%. The tightest reported setting, C2, is qualitatively different: Phase 1 still solves the relaxed problem quickly, but Phase 2 repeatedly separates feasibility cuts and reaches the time limit. Thus, the complete method handles the tested C3–C6 range, while C2 exposes a clear limitation: sufficiently tight shared capacity can make exact verification the computational bottleneck.

6. Conclusion

This paper studies wireless charging infrastructure planning for electric-bus systems under uncertain future service plans. The two-stage formulation separates here-and-now SWC and DWC deployment from scenario-adaptive route battery configuration and representative-day charging decisions. Its recourse combines scenario-dependent discrete configurations, route-level energy feasibility, and shared plug-in charger capacity. To exploit this structure, we develop an exact integer L-shaped method that lifts the battery configurations into the decomposition master, uses route-level LP recourse to generate lower bounds before exact scenario verification, and strengthens scenario cuts through a joint facility–battery resource order. The resulting procedure preserves the original information structure and converges finitely under exact separation.

The Tower-derived experiments identify the two-phase relaxation–verification scheme as the main computational improvement in the tested method ladder. The two-phase variants solve the small, medium, and base three-scenario instances, whereas the standard integer L-shaped variants do not solve even the small instance within the two-hour limit. On the base network, the full method solves all cases from 3 to 40 scenarios; the 40-scenario instance terminates in 926.4 seconds with a final gap below 0.004%. The boundary cases also clarify where the method remains expensive: the largest three-scenario network reaches the time limit at a 0.092% gap, and the tightest plug-in capacity setting fails to complete exact Phase 2 verification. Thus, after route-level lower bounding reduces the cost of early separation, exact verification becomes the principal bottleneck when the network is large or shared-capacity coupling is tight.

The study uses constructed service-plan scenarios on a fixed route network, treats existing plug-in charger capacity as exogenous, and assumes that trip energy consumption is independent of the selected battery capacity. Future work could develop data-driven service-plan scenarios, jointly optimize wireless and plug-in capacity expansion, and incorporate battery-weight effects and short-term operational uncertainty. From an algorithmic perspective, stronger exact-verification procedures for tightly coupled recourse and tests on other stochastic design problems would help establish how broadly the layered decomposition applies.

Appendix A. Model Notation

Appendix B. Proofs

Appendix B.1. Route-Level LP Cuts for Phase 1

For a fixed scenario–route pair (ω, r), express the Phase 1 route LP (25) in the equivalent parametric form

$$Q^R_{\omega,r}(w, z_{\omega,r}) = \min_{y_{\omega,r}\geq 0} \left\{ c^\top_{\omega,r} y_{\omega,r} : A_{\omega,r} y_{\omega,r} \geq b_{\omega,r} + H_{\omega,r} w + G_{\omega,r} z_{\omega,r} \right\}.$$

Here, $c_{\omega,r}$, $A_{\omega,r}$, $b_{\omega,r}$, $H_{\omega,r}$, and $G_{\omega,r}$ collect the corresponding fixed objective and constraint coefficients. If the route LP is feasible, any dual-feasible vector

$$\pi_{\omega,r} \geq 0, \qquad A^\top_{\omega,r} \pi_{\omega,r} \leq c_{\omega,r},$$

Table A.8: Summary of notation

Symbol	Description
Sets and indices	
Ω	Set of service-plan scenarios, indexed by ω
$\mathcal{S}$	Set of stations, indexed by s
$\mathcal{L}$	Set of directed links between consecutive stations, indexed by l
C	Set of existing interchanges or hubs equipped with plug-in chargers, $C \subseteq \mathcal{S}$
$\mathcal{R}$	Set of routes, indexed by r
$\mathcal{K}$	Set of candidate battery-capacity options, indexed by k
$\mathcal{M}_{\omega,r}$	Set of bus trip chains on route r under scenario ω, indexed by m
$\mathcal{V}_{\omega,r}$	Set of station-visit event copies generated by all trip chains on route r under scenario ω, indexed by v
$\mathcal{V}^{\mathrm{P}}_{\omega,r}$	Set of visit event copies eligible for plug-in charging on route r under scenario ω
$\mathcal{V}^{\mathrm{S}}_{\omega,r}$	Set of visit event copies eligible for SWC on route r under scenario ω
$\mathcal{A}_{\omega,r}$	Set of directed-traversal event copies generated by all trip chains on route r under scenario ω, indexed by a
$\mathcal{T}_{\omega}$	Relevant plug-in charging time periods under scenario ω
$\mathcal{T}_{v}$	Time periods during which plug-in charging visit v can occupy a plug-in charger
Parameters	
p_{ω}	Probability of scenario ω
c^{S}_{s}	Representative-day equivalent cost of installing SWC at station s
c^{D}_{l}	Representative-day equivalent cost of installing DWC on link l
c^{bat}	Representative-day equivalent battery capacity cost per kWh
$c^{\mathrm{day}}, c^{\mathrm{night}}$	Daytime and overnight electricity prices
$n_{\omega,r}$	Fleet-size requirement for route r under scenario ω
β_k	Battery capacity value represented by option k
d_a	Energy consumption associated with directed link traversal a
d^{0}_{m}	Boundary energy consumption from the depot to the first station visit of trip chain m
d^{N}_{m}	Boundary energy consumption from the terminal station visit of trip chain m back to the depot
U_s	Number of existing plug-in chargers at station s
κ	Grid-side energy delivered by one occupied plug-in time slot
$\bar{q}^{\mathrm{S}}_{v}$	Maximum SWC energy available during station visit v
$\bar{q}^{\mathrm{D}}_{a}$	Maximum DWC energy available during directed link traversal a
$\eta^{\mathrm{P}}, \eta^{\mathrm{S}}, \eta^{\mathrm{D}}$	Charging efficiencies for plug-in charging, SWC, and DWC
$\underline{\rho}, \overline{\rho}$	Minimum and maximum state-of-charge fractions
Derived quantities and functions	
$s(v)$	Physical station associated with visit event v
$\ell(a)$	Physical directed link associated with traversal event a
$\mathrm{tail}(a)$	Departure visit event of traversal event a
$\mathrm{head}(a)$	Arrival visit event of traversal event a
v^{0}_{m}	First visit event of trip chain m
v^{N}_{m}	Last visit event of trip chain m
Decision variables	
w^{S}_{s}	1 if SWC is installed at station $s \in \mathcal{S} \setminus C$, 0 otherwise
w^{D}_{l}	1 if DWC is installed on link $l \in \mathcal{L}$, 0 otherwise
$z_{\omega,r,k}$	1 if route r uses battery capacity option k in scenario ω, 0 otherwise
$B_{\omega,r}$	Selected battery capacity for route r in scenario ω
x_{ω}	Vector of scenario-level plug-in charger occupancy decisions in scenario ω
y_{ω}	Vector of scenario-level charging amounts and battery energy-state decisions in scenario ω
q^{P}_{v}	Plug-in charging energy assigned during plug-in visit v
q^{S}_{v}	SWC energy assigned during SWC-eligible station visit v
q^{D}_{a}	DWC energy assigned to directed link traversal a
q^{N}_{m}	Overnight charging energy assigned after trip chain m returns to the depot
$e^{\mathrm{arr}}_{v}, e^{\mathrm{dep}}_{v}$	Battery energy state at arrival and departure of station visit v
$x_{v\tau}$	1 if plug-in charging visit v occupies one plug-in charger at time τ, 0 otherwise

gives the route-level optimality cut

$$\theta^R_{\omega,r} \geq \pi^\top_{\omega,r} \left(b_{\omega,r} + H_{\omega,r} w + G_{\omega,r} z_{\omega,r}\right).$$

If the route LP is infeasible at an evaluated configuration $(w^*, z^*_{\omega,r})$, a Farkas ray can be selected from

$$\Sigma^{\mathrm{F}}_{\omega,r} = \left\{\sigma_{\omega,r} \geq 0 : A^\top_{\omega,r} \sigma_{\omega,r} \leq 0\right\}$$

such that

$$\hat{\sigma}^\top_{\omega,r} \left(b_{\omega,r} + H_{\omega,r} w^* + G_{\omega,r} z^*_{\omega,r}\right) > 0.$$

Then any configuration with a feasible Phase 1 route relaxation must satisfy

$$\hat{\sigma}^\top_{\omega,r} \left(b_{\omega,r} + H_{\omega,r} w + G_{\omega,r} z_{\omega,r}\right) \leq 0. \tag{B.1}$$

Thus (B.1) is a valid feasibility cut for the Phase 1 relaxation.

Appendix B.2. Finite convergence of the two-phase method

Proof of Theorem 1. Proposition 1 and cut validity make every Phase 2 restricted master a relaxation of the exact lifted formulation, so its optimal value is a valid lower bound. Whenever all exact scenario subproblems are proven feasible and solved to optimality, (32) gives a valid upper bound. At any Phase 2 master configuration, exact separation either excludes an infeasible scenario configuration or enforces its exact recourse value whenever $\theta_\omega < Q_\omega(w, z_\omega)$. Because the binary configuration space and scenario set are finite and all corrections are retained, only finitely many such corrections can occur.

At the final uncorrected master solution, every scenario is exact-feasible. If $\theta_\omega < Q_\omega(w, z_\omega)$ for any scenario, exact separation would add a further correction; hence $\theta_\omega \geq Q_\omega(w, z_\omega)$ at termination. All retained constraints on θ_ω are valid lower bounds on $Q_\omega(w, z_\omega)$, so decreasing any θ_ω above $Q_\omega(w, z_\omega)$ preserves master feasibility. Because $p_\omega > 0$ and the master minimizes θ_ω, no optimal master solution can choose such an excessive value, and therefore $\theta_\omega = Q_\omega(w, z_\omega)$ for every scenario. The master objective then equals the exact candidate value, so $LB = UB$ when $\varepsilon = 0$. Exactness of the lifted reformulation completes the result. □

Appendix C. Additional Computational Results

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